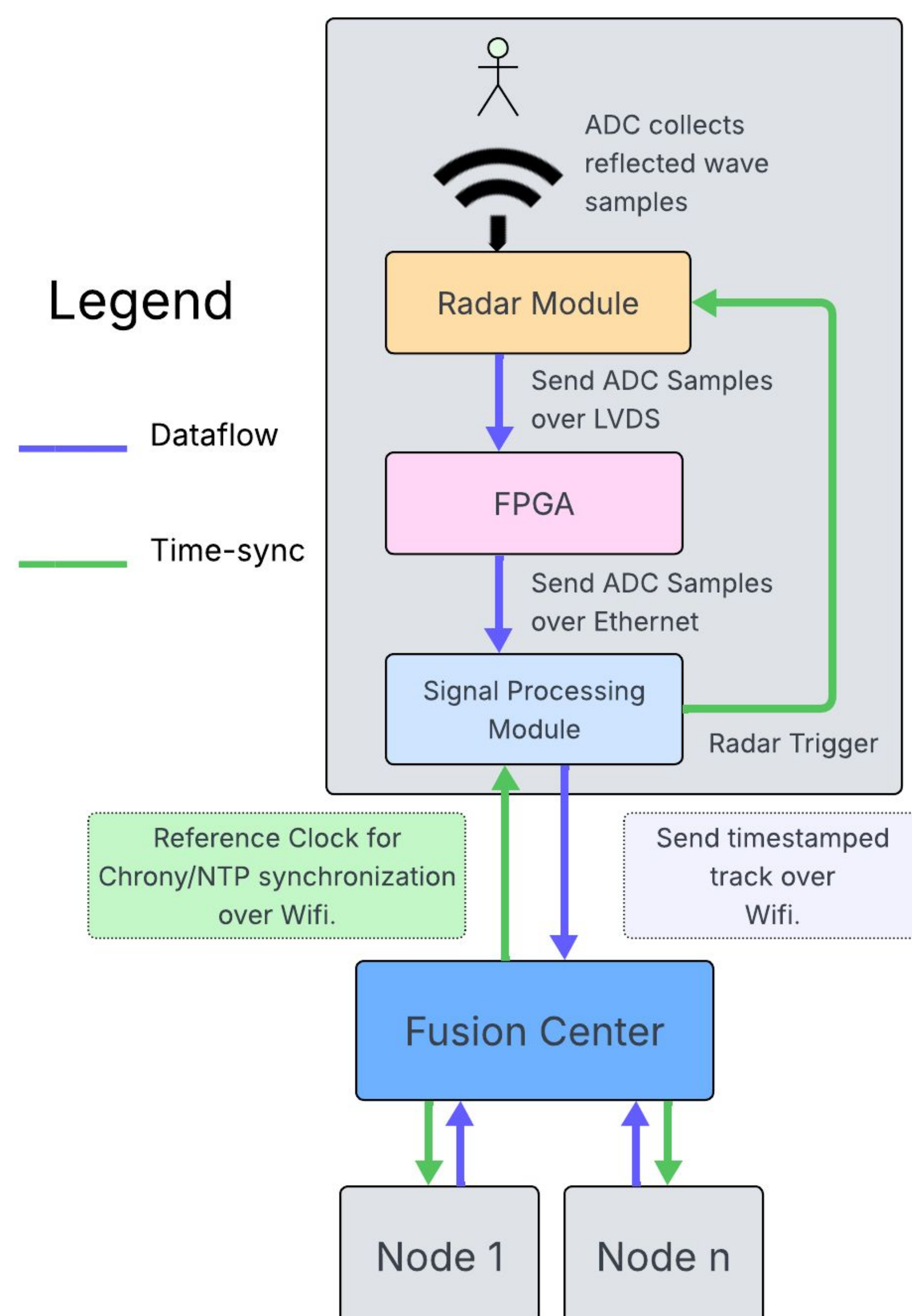


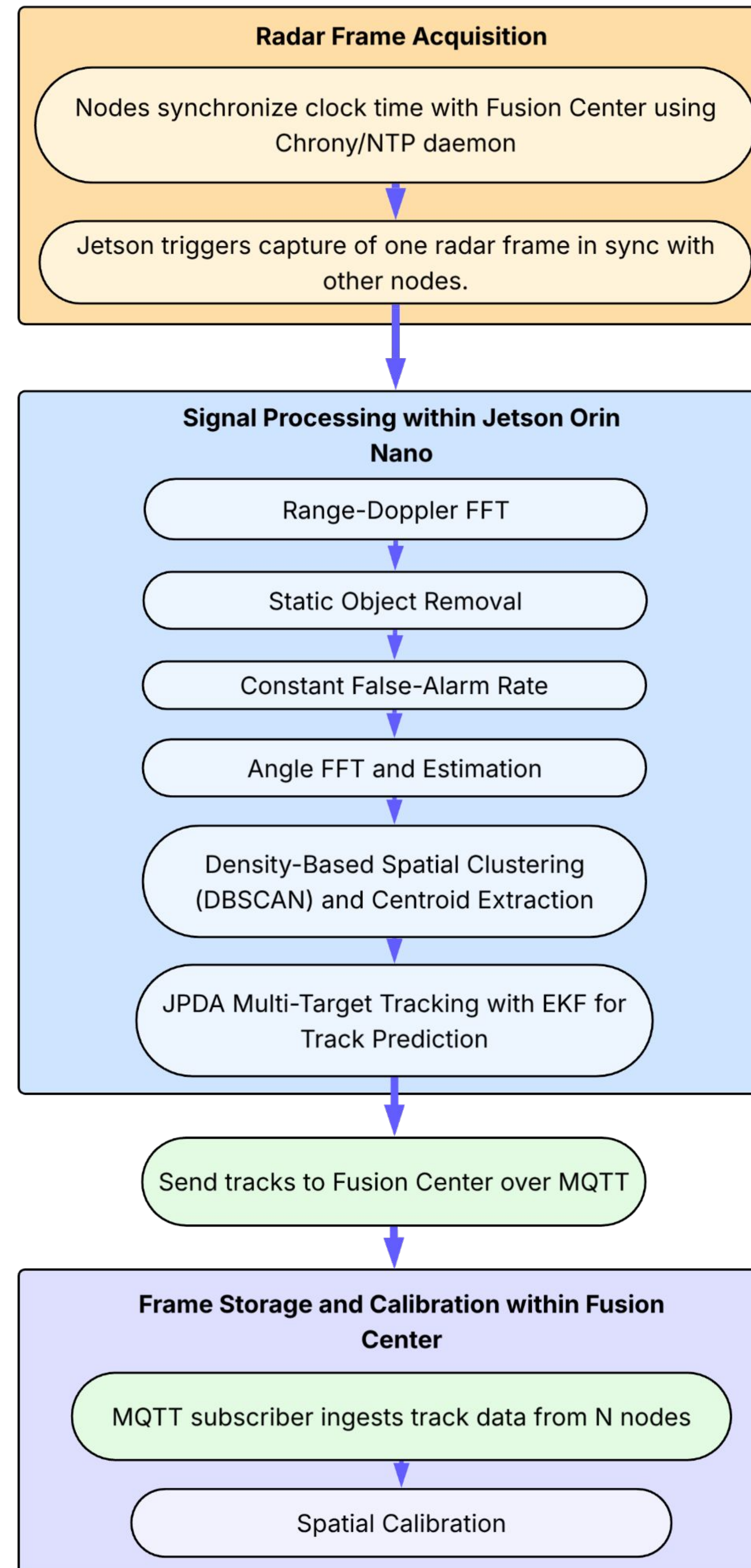
Background

Object tracking using mmWave MIMO radar is well-established, but sensing performance degrades in cluttered environments. We present a time-synchronized distributed radar network that fuses detections across spatially separated nodes, enabling continuous tracking through multiple fields of view and line-of-sight obstructions. This architecture extends spatial coverage and maintains track continuity in scenarios where single-node radar systems fail.

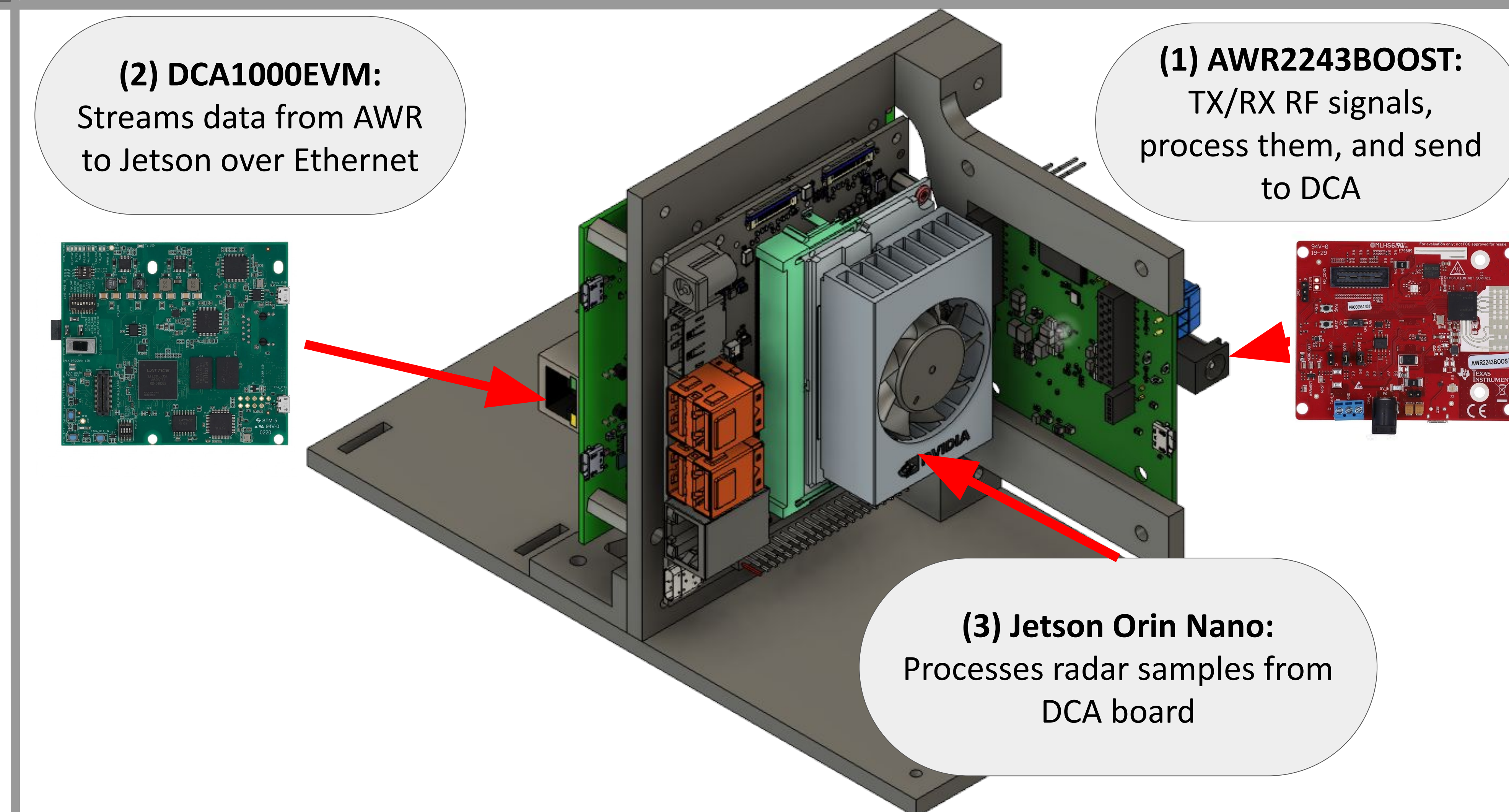
Block Diagram



Data Flow

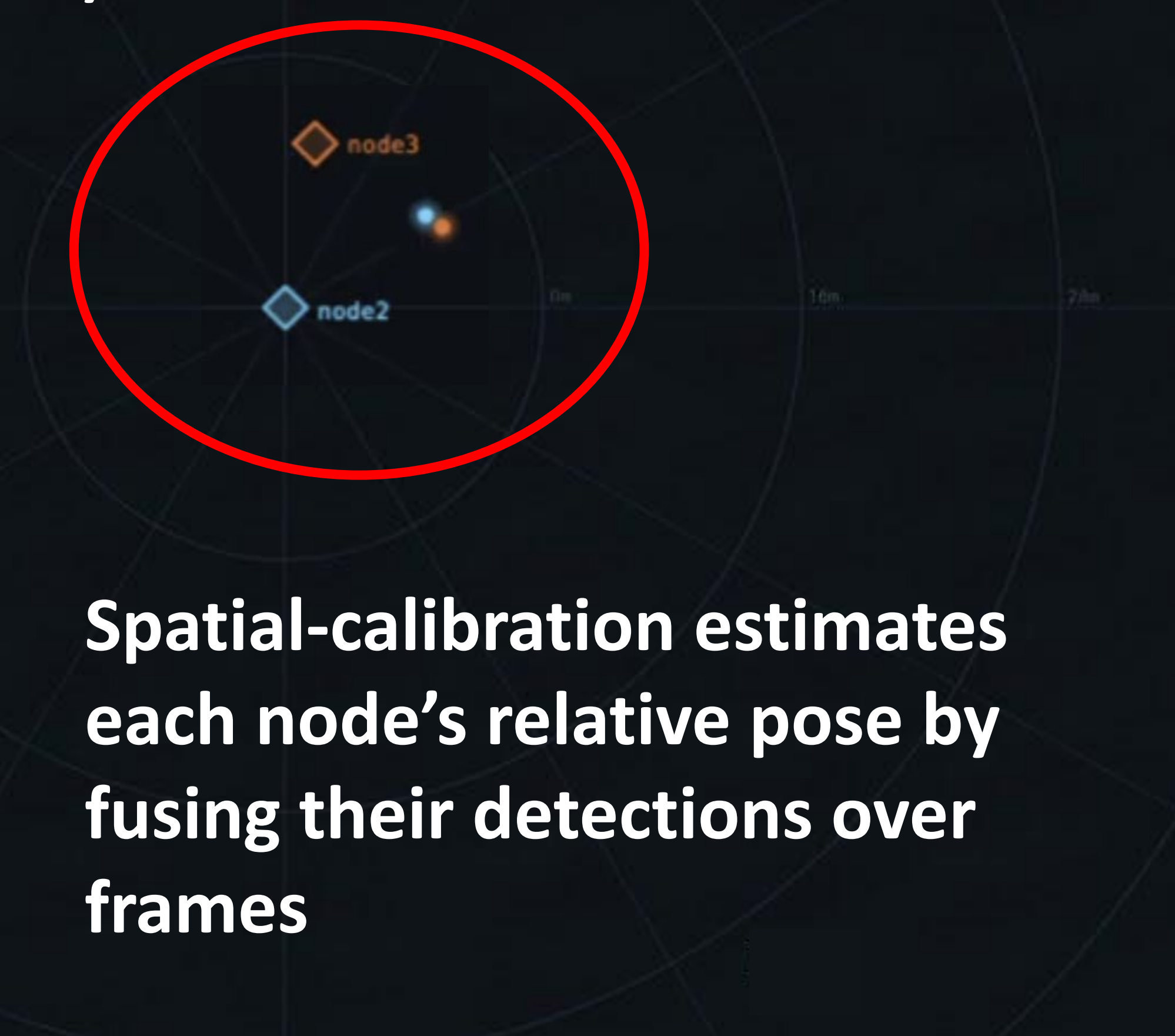


Radar Node



Birds-Eye Visualizer

Live demonstration of Vihan walking through the overlapping FOV of 2 radar nodes facing each other. The visualizer shows each node (denoted as a diamond), and each node's centroid detections (denoted as a solid dot)



Spatial-calibration estimates each node's relative pose by fusing their detections over frames

Live demonstration of Vihan walking through the overlapping FOV of 2 radar nodes facing each other. The visualizer shows each node (denoted as a diamond), and each node's centroid detections (denoted as a solid dot)



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Spatial-calibration estimates each node's relative pose by fusing their detections over frames





Chirp

(OLD DRAFT)

Distributed Radar Network

William Ni | Oviya Seeniraj | Vihan Jayaraman | Andrey Otvagin | Jason Wang

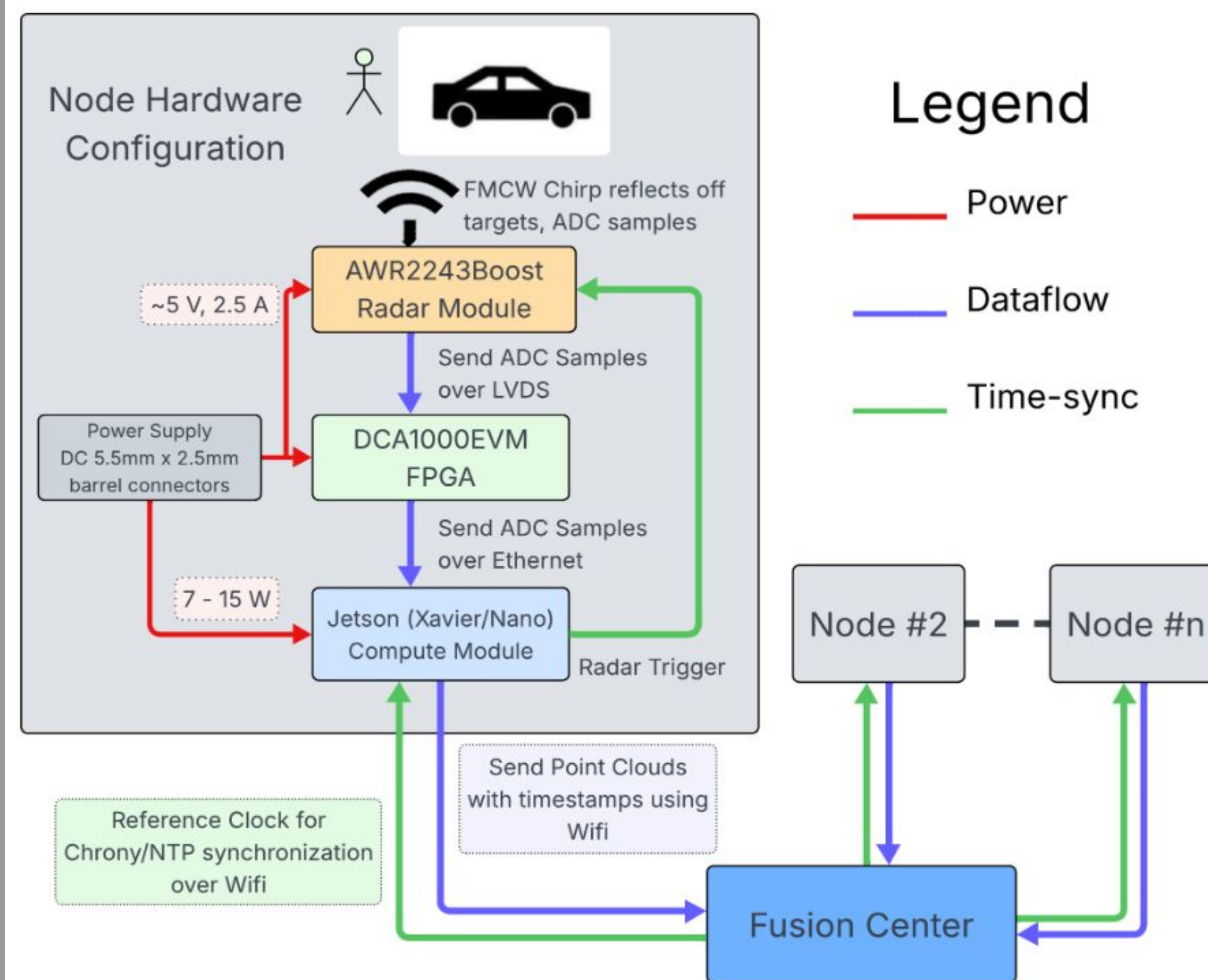
Background

Individual commercial mmWave MIMO radar nodes can track objects within their limited field of view (FOV), but their sensing capability is confined to a small area. When multiple radar nodes are organized into a distributed network, they can collaborate to jointly track targets across a broader region, expanding coverage and improving situational awareness.

Overview

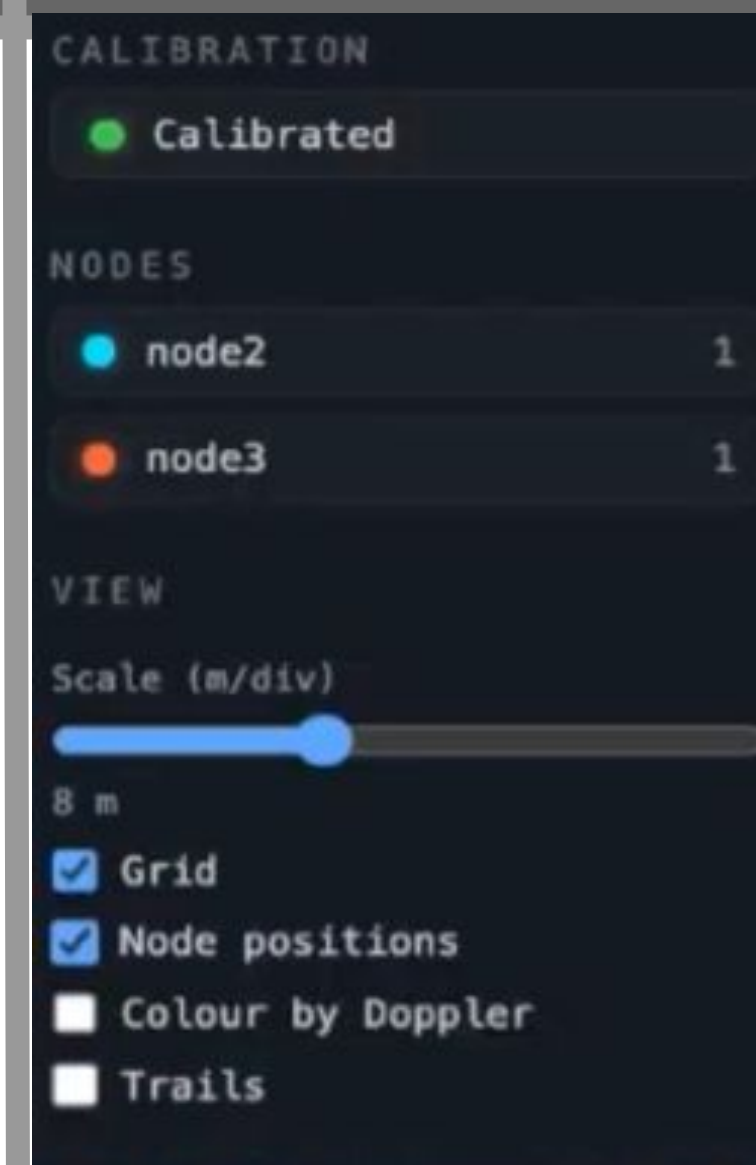
This work presents a networked radar sensing framework that synchronizes node clocks through a local network connection. Once synchronized, nodes coordinate with a central server to simultaneously capture radar frames, detect shared targets within overlapping FOVs, and process signals into centroids containing range, Doppler, and angle information. These centroids are transmitted to a fusion center, which applies an auto-calibration algorithm to estimate the relative position and orientation of each radar node.

Block Diagram



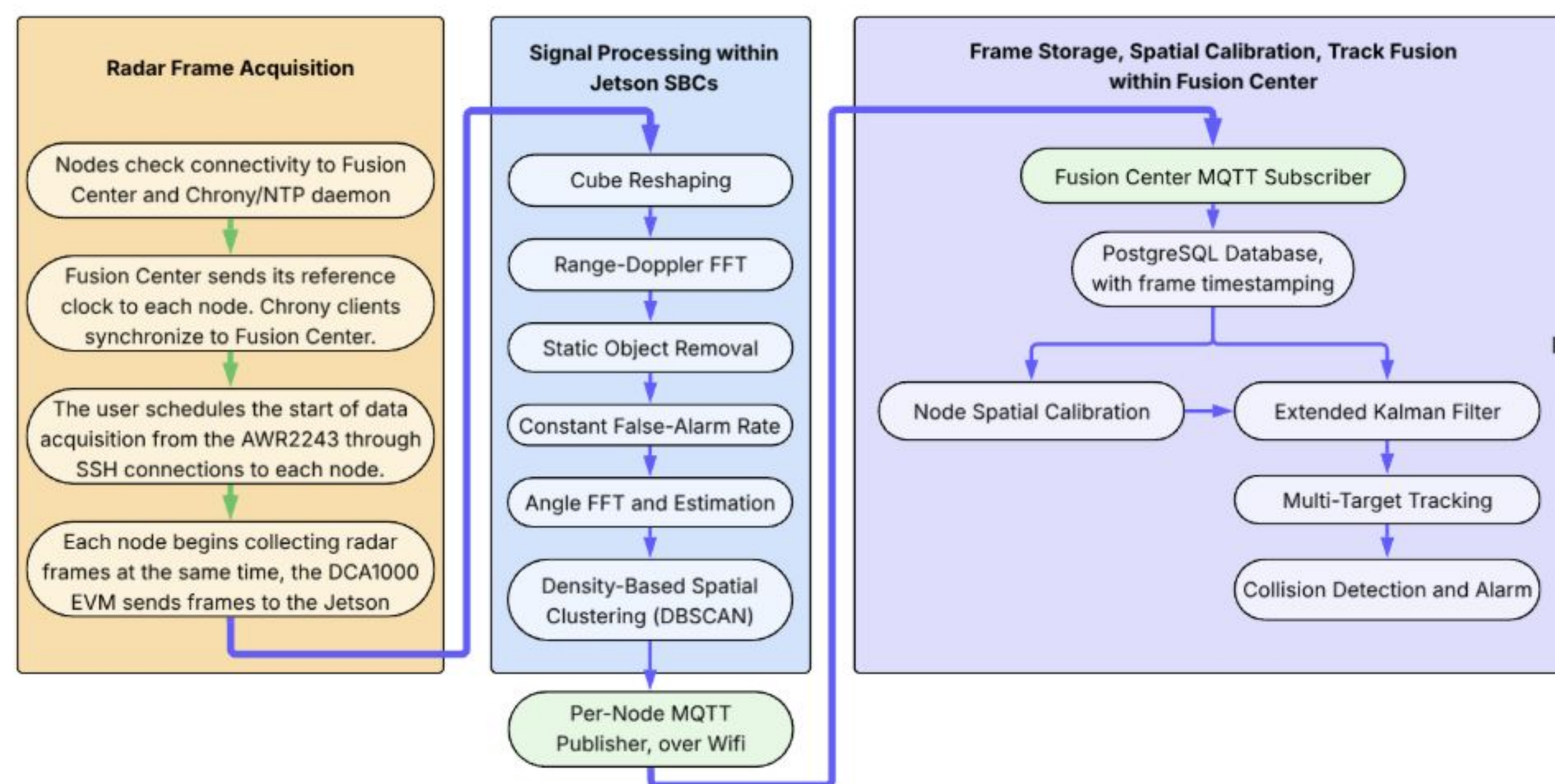
Birds-Eye Visualizer

Live demonstration of Vihan walking across the overlapping FOV of 2 radar nodes facing each other. The visualizer shows each node (denoted as a diamond), and each node's centroid detections (denoted as a solid dot).

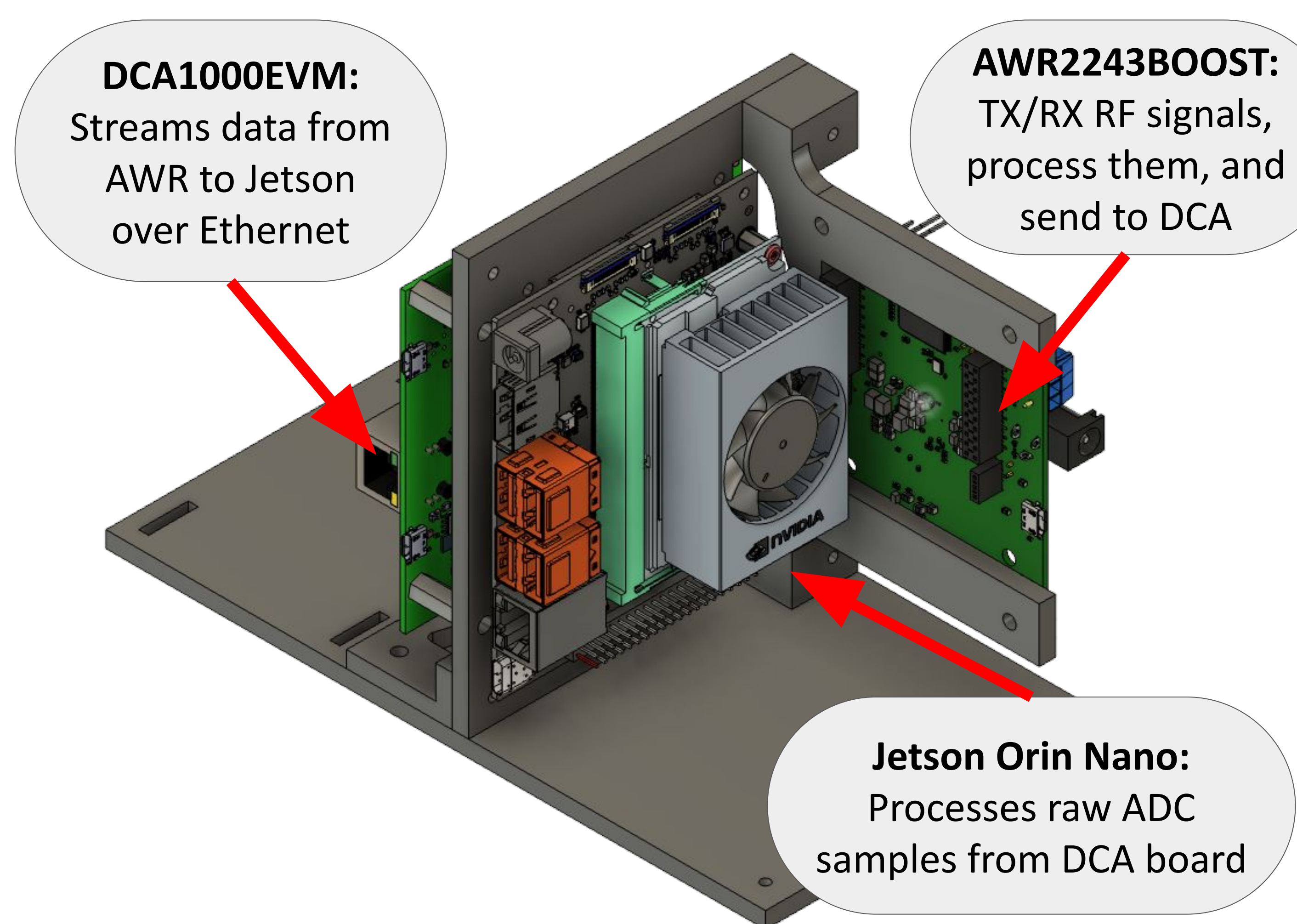


Auto-calibration estimates each node's relative pose by fusing their detections over frames.

Data Flow



Radar Node



Acknowledgements: